

Vedansh Somani

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🐙 github.com/sudoVed 🔗 Portfolio - vhades.dpdns.org

PROFILE

Robotics & AI undergraduate interested in building intelligent, autonomous systems through reinforcement learning and multi-agent learning. Seeking research and engineering opportunities focused on challenging real-world AI and robotics problems.

EDUCATION

Bachelor of Engineering - Robotics and Artificial Intelligence | CGPA: 9.07 09/2023 – Present
Dayananda Sagar College of Engineering Bengaluru, India

Central Board of Secondary Education- Science | Percentage: 94.4% 2021 – 2023
Sai International Residential School Cuttack, India

SKILLS

Machine Learning

PyTorch, Reinforcement Learning, Multi-Agent RL, Deep Learning

Robotics and Autonomous Systems

ROS2, SLAM, Nav2, MuJoCo, Path Planning, Robot Control, Autonomous Navigation

Systems and Programming

Python, C++, Linux, Git, WebAssembly, SIMD, Embedded Systems

Research Interests

Reinforcement Learning, Multi-Agent Systems, Robot Learning, AI for IoT

PROJECTS

Emergent Communication in Multi-Agent Reinforcement Learning 08/2026 – Present
Multi-Agent RL, MAPPO, PyTorch, Emergent Communication

Investigating **MAPPO-based emergent communication** where cooperative agents learn what, when, and with whom to communicate, benchmarked against no-communication and fixed-communication baselines for task performance, efficiency, and robustness.

STRIDE v1 — Quadruped Locomotion via Reinforcement Learning 05/2026 – 06/2026
Python, MuJoCo, PPO, PyTorch

Trained a single **PPO locomotion policy** for a 15 kg Unitree Go2 using **PMTG gait generation** and **residual PD control**, achieving 100% recovery at 90 N on flat terrain and 91.7% on 0.15 m uneven terrain.

Chess Engine — NNUE Evaluation Engine 06/2026 – 07/2026
C++, WebAssembly, SIMD, PyTorch

Built a C++ alpha-beta chess engine and trained a **HalfKP NNUE evaluator**, using incremental accumulators, integer quantization, and WASM SIMD to cut evaluation cost $\sim 3-4\times$ and reach depth 10 in ~ 330 ms.

Chain Reaction — Fully Convolutional DQN Agent 04/2026 – 05/2026
Python, PyTorch, Deep Q-Learning

Trained a **fully convolutional DQN** with 4-channel spatial board encoding, curriculum learning, reward shaping, and action masking, achieving 95% / 81.5% / 57% win rates vs. random / defensive / greedy opponents.

Autonomous Mobile Robot 02/2026 – 05/2026
ROS2, Nav2, SLAM, LiDAR

Developed a **differential-drive AMR** on ROS2 + Raspberry Pi 5, integrating LiDAR odometry/SLAM, Nav2 autonomous navigation, and a custom ROS2-to-Arduino motor bridge for real-time control.